

CROSSVILLE, TN, USA

WMO: 723250

Lat: 35.951N

Lon: 85.081W

Elev: 1867

StdP: 13.73

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03847

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.4	15.9	0.5	6.0	17.0	5.1	7.7	19.3	20.1	44.5	18.3	43.8	4.9	320	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	88.0	72.0	85.7	71.5	83.9	71.0	75.1	83.1	74.0	81.9	73.0	80.6	6.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	128.9	79.4	71.7	125.4	78.5	70.4	120.1	77.1	39.7	83.3	38.7	82.0	37.7	80.7	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.0	15.0	12.8	DB	2.9	90.9	7.1	3.6	-2.3	93.4	-6.4	95.5	-10.5	97.5	-15.6	100.1	
			WB	2.0	76.8	6.9	1.8	-2.9	78.1	-7.0	79.1	-10.8	80.2	-15.8	81.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.5	36.1	39.9	47.3	56.8	64.4	71.2	74.3	73.8	67.9	57.6	47.2	39.9
	DBStd	15.79	11.95	10.78	10.22	8.68	6.89	4.50	3.59	3.93	5.88	8.15	9.25	10.36
	HDD50	1501	452	308	173	36	3	0	0	0	0	27	160	342
	HDD65	4124	897	702	550	264	96	9	0	1	39	251	537	778
	CDD50	3858	20	26	90	240	449	637	755	737	539	262	74	29
	CDD65	1006	0	0	2	19	77	196	290	273	128	20	1	0
	CDH74	6970	0	0	10	128	508	1336	2030	1957	876	121	4	0
	CDH80	1699	0	0	0	6	62	309	546	559	204	13	0	0
Wind	WSAvg	5.1	6.6	6.5	6.7	6.5	5.0	4.0	3.6	3.3	3.5	4.4	5.5	6.2
Precipitation	PrecAvg	54.5	4.6	4.4	5.8	4.8	5.3	4.4	5.1	4.0	3.8	3.0	4.8	5.3
	PrecMax	69.6	10.1	10.5	15.4	11.3	11.7	14.1	12.0	9.0	9.0	7.2	8.5	12.8
	PrecMin	38.5	0.8	0.8	1.4	1.6	1.6	0.9	1.4	0.9	0.7	0.0	1.7	0.9
	PrecStd	9.1	2.4	1.9	3.2	2.5	2.3	2.8	2.6	2.0	2.2	1.9	1.9	2.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.2	67.7	75.2	80.9	84.2	88.3	90.4	91.2	88.3	81.2	73.0	67.6
		MCWB	57.6	58.5	59.9	62.3	68.7	72.0	73.2	72.0	68.2	66.3	61.0	60.7
	2%	DB	61.1	63.8	71.6	77.7	81.8	85.6	87.6	88.0	84.7	77.2	68.7	63.3
		MCWB	56.3	55.9	57.4	61.0	66.8	71.4	73.1	72.3	68.5	64.2	57.8	57.5
	5%	DB	57.7	60.6	67.3	74.8	79.5	83.6	85.0	85.3	82.2	74.3	65.2	59.5
		MCWB	53.1	53.0	55.7	59.9	65.8	70.6	72.6	71.3	68.2	62.1	57.1	54.8
	10%	DB	54.2	56.4	63.6	71.8	76.9	81.4	83.1	83.0	79.4	71.3	62.3	55.7
		MCWB	50.1	49.7	53.9	58.5	64.7	69.8	72.0	71.2	67.4	61.3	55.6	51.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.3	61.5	62.5	66.8	71.6	75.3	76.8	76.4	73.0	70.0	64.3	62.4
		MCDB	63.3	65.5	69.7	75.8	79.9	83.3	84.9	85.0	80.7	76.6	69.0	65.8
	2%	WB	57.3	58.7	60.3	64.5	69.5	73.8	75.3	74.7	71.4	67.5	61.9	59.0
		MCDB	60.3	62.8	67.5	73.0	77.6	82.0	83.0	83.2	79.3	73.7	66.2	62.4
	5%	WB	54.3	55.1	58.3	62.7	68.2	72.6	74.3	73.5	70.4	65.6	59.3	55.6
		MCDB	57.2	58.2	65.2	70.4	75.9	80.4	82.0	81.5	77.9	71.0	63.5	58.7
	10%	WB	50.8	51.2	55.8	60.8	66.9	71.2	73.3	72.4	69.1	63.6	56.4	52.1
		MCDB	53.3	55.1	62.1	68.3	74.1	78.5	80.8	80.0	76.1	68.4	60.8	55.3
Mean Daily Temperature Range	5% DB	MDBR	16.5	17.1	19.0	20.2	19.4	18.1	16.7	17.8	19.3	20.4	18.8	15.9
		MCDBR	17.0	20.2	22.6	22.9	21.6	20.9	18.9	20.7	22.7	23.0	21.3	17.8
		MCWBR	12.8	13.6	12.9	11.1	9.7	9.1	8.0	8.3	9.3	11.6	12.5	13.1
	5% WB	MCDBR	14.3	16.3	18.5	18.4	17.8	17.6	16.5	17.0	17.5	16.8	16.0	14.9
		MCWBR	12.5	13.9	12.5	10.5	9.1	8.8	7.9	8.0	8.4	10.2	12.0	13.2
Clear-Sky Solar Irradiance	taub	0.300	0.312	0.339	0.376	0.421	0.452	0.481	0.490	0.396	0.353	0.317	0.297	
	taud	2.524	2.480	2.440	2.326	2.225	2.166	2.114	2.074	2.323	2.441	2.506	2.548	
	Ebn at Noon	287	295	295	288	275	265	256	251	272	276	278	281	
	Edh at Noon	25	30	34	40	45	48	50	51	37	30	26	23	
All-Sky Solar Radiation	RadAvg	705	883	1249	1626	1801	1947	1842	1724	1462	1159	835	609	
	RadStd	62	113	122	114	143	127	105	88	139	149	93	59	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+157	+117
(47)	Regional (24 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+247	+166

Nomenclature: See separate page